Relationships between variables

1

a Yes **b** No **c** No **d** Yes

2

a No.

b Distance from the equator affects temperature.

c No.

d Distance covered affects time of travel or time of travel affects distance covered.

e No

f Time spent working affects wages earned.

g Number of pages affects time spent to finish a novel.

h No

i No.

j No

k Number of pets affects time spent caring for pets (hours).

l Time spent training (hours) affects athletic performance.

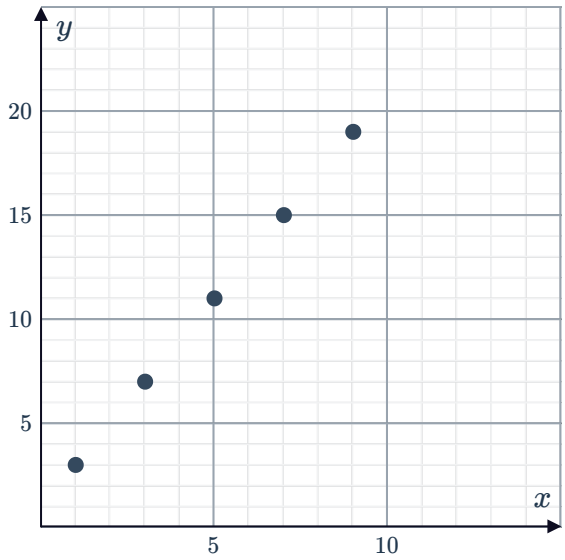
3 Negative

Scatter plots

4

a Yes **b** No **c** No

5



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6

a Strong positive correlation

b No correlation

c Strong negative correlation

d Weak positive correlation

e Weak negative correlation

7 There is no direct relationship

8

a As air temperature increases, sea temperature increases.

b Strong, positive linear correlation

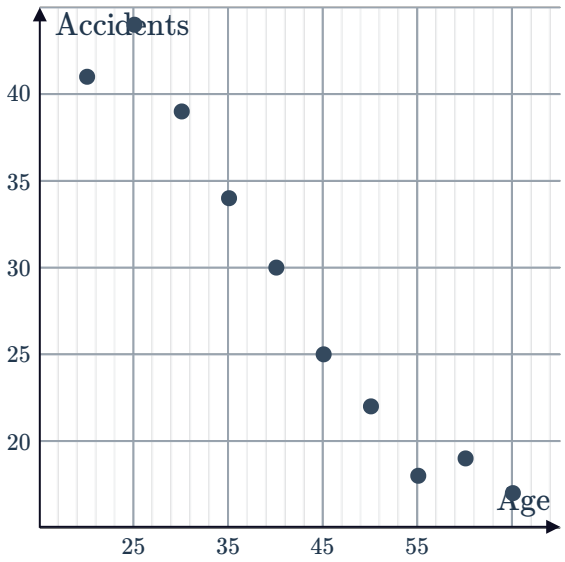
9

a Amount of healthy coral

b Sea temperature

10

a



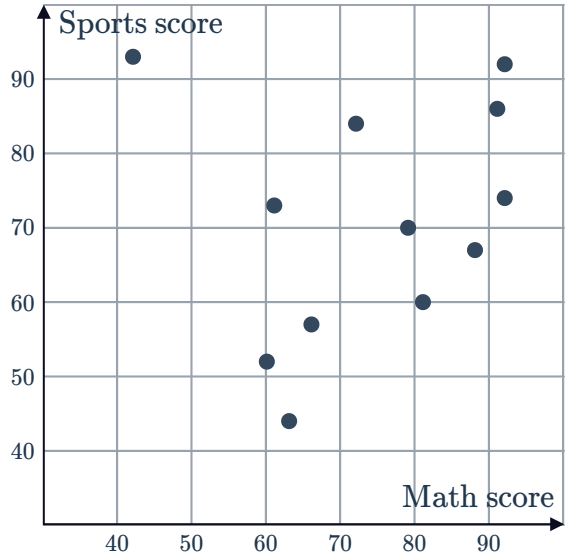
b Strong, negative, linear

c No

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11

a

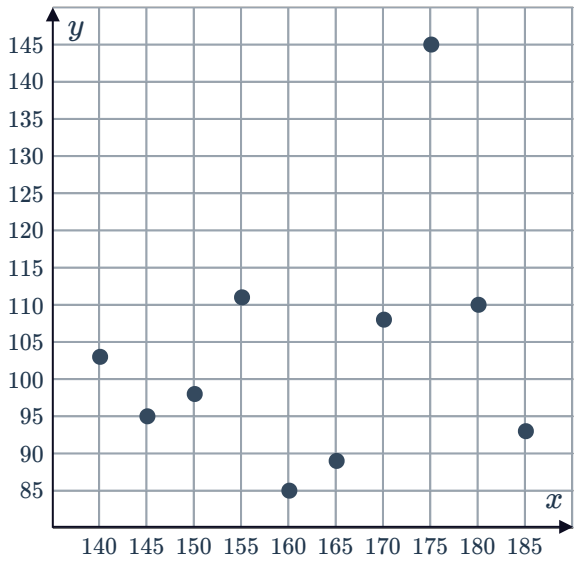


b Weak, positive, linear

c Student 10's

12

a



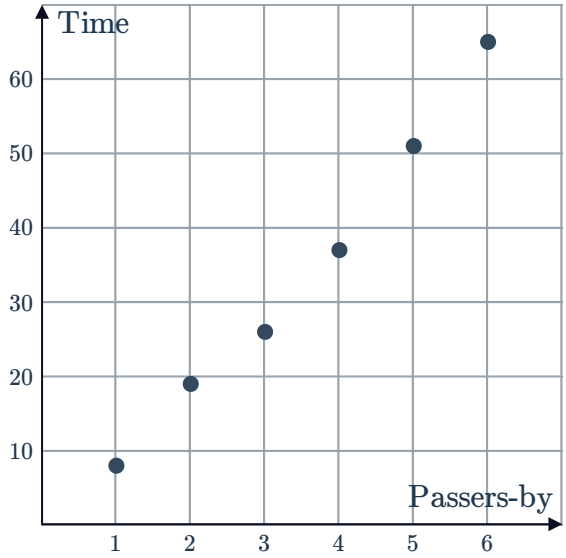
b Not correlated

c Height $= 175$ cm

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13

a

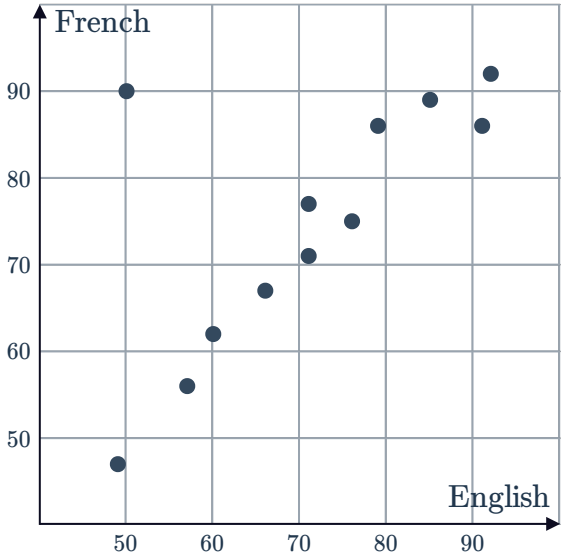


b The more passers-by there are, the longer will be the time until help is offered.

c Strong, positive, linear

14

a



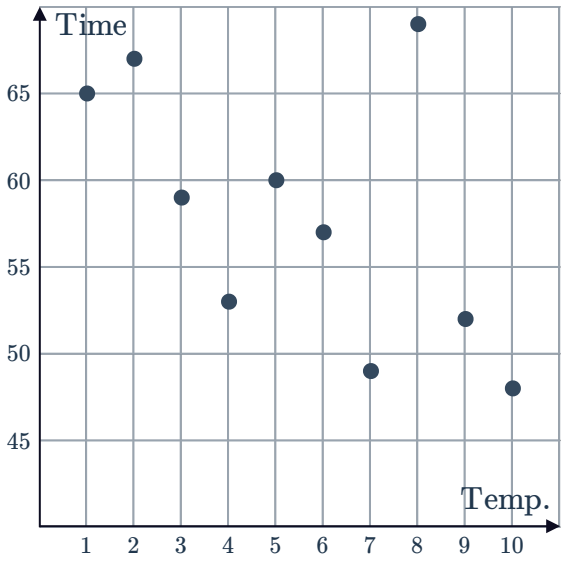
b Strong, positive, linear

c Student 9's

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15

a



b 10

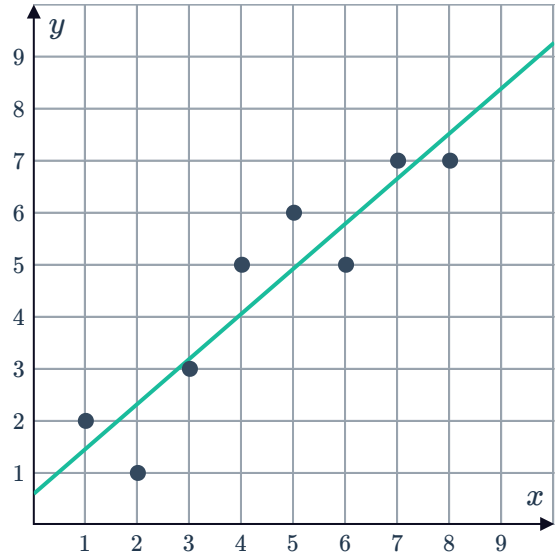
c Moderate, negative, linear

d 8˚C, 69 seconds

Lines of best fit

16

a



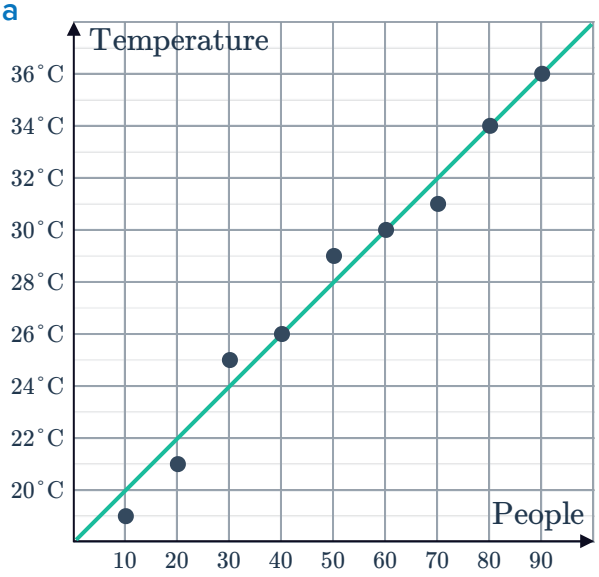
b

i Approx. 4.5 **ii** Approx. 8.4

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17

a



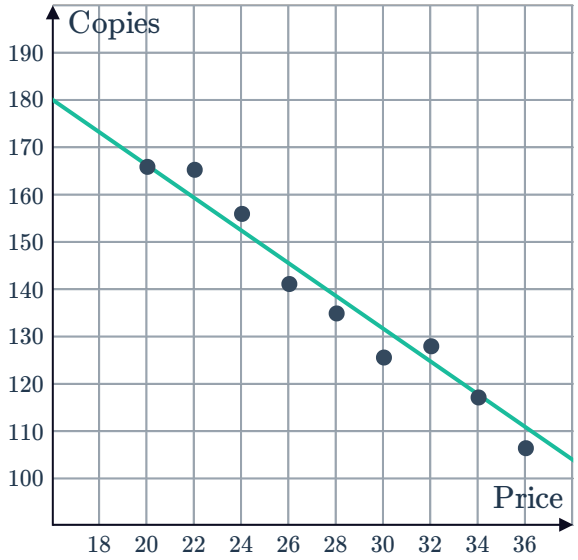
b

i Approx. 25°C

ii Approx. 38°C

18

a

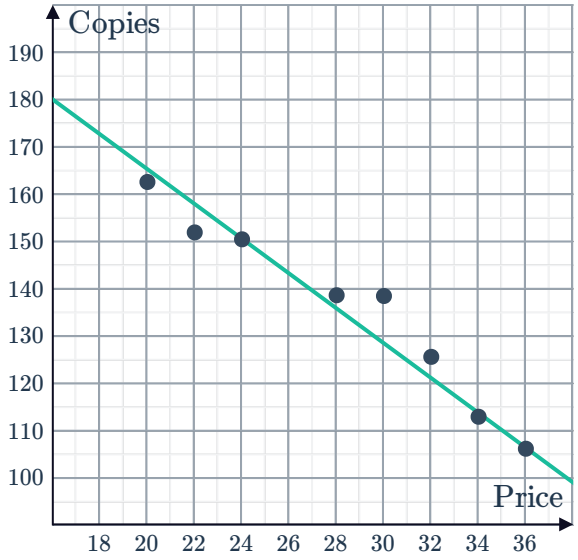


b

i Approx. 121 ii Approx. 174

19

a



b

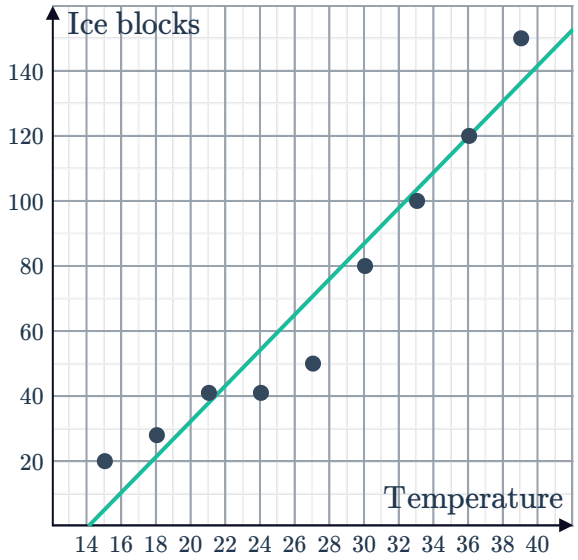
i Approx. 117 ii Approx. 173

c Negative

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20

a



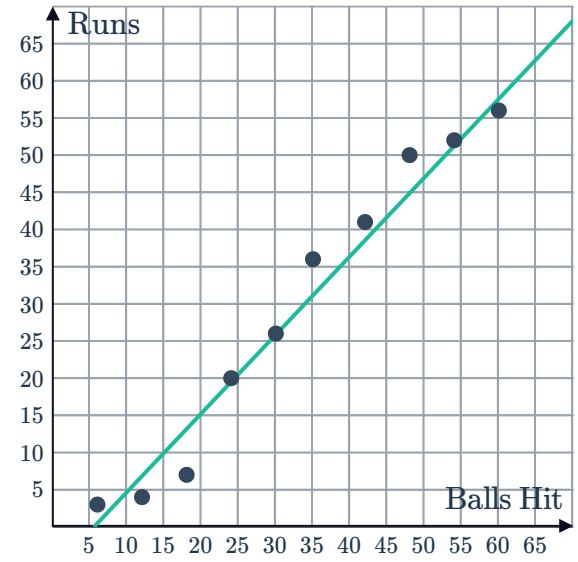
b

i Approx. 91 **ii** Approx. 152

c Increases

21

a



b

i Approx. 23 **ii** Approx. 63

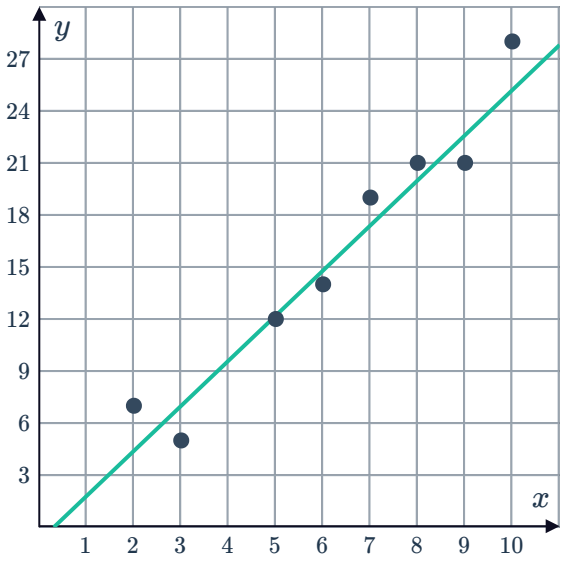
c Positive

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22

a Positive

b



c **D**

23

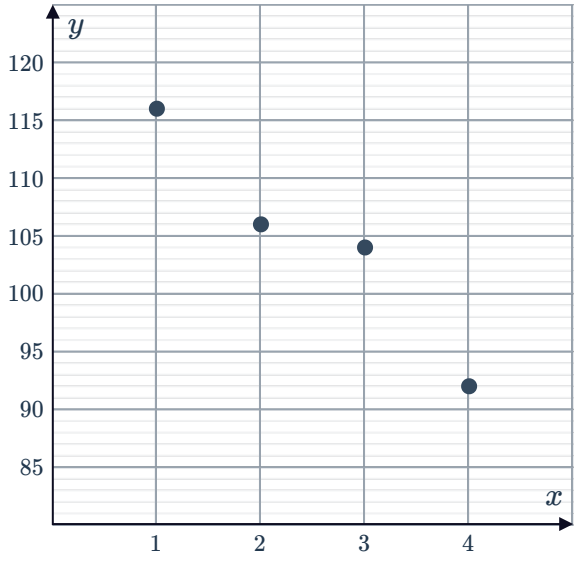
a Negative

b **C**

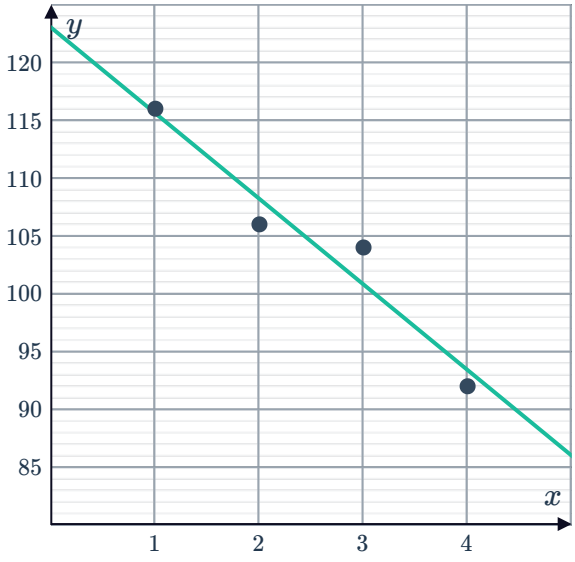
24 4 knots

25

a



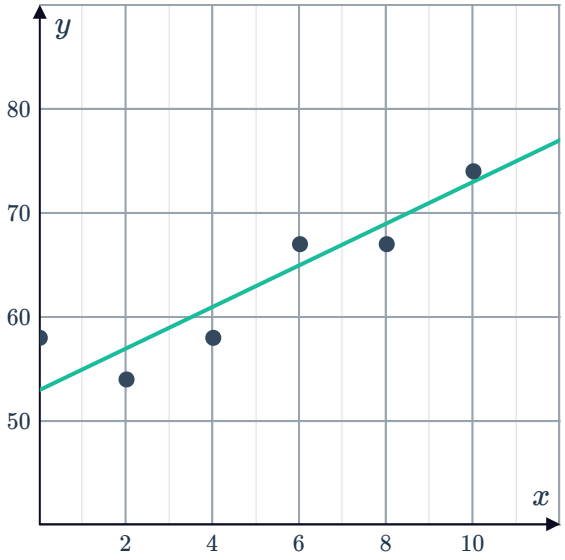
b



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26

a



b $(10, 74)$

c $(0, 58)$

27

a 20 years **b** 1300 fish **c** 11 years

28

a 3 seconds **b** 2.5 hours

29

a 20°C **b** 210 chirps per minute

c 130 chirps per minute

30

a 8500 m **b** 6500 m **c** 300 seconds **d** 1800 seconds

31

a 90. When a student watched no TV, their average result in the exam was 90.

b Yes. When a student watches 0 hours, their time is within the data range, and the result makes sense.

c No

32

a 30

b When a child doesn't read any pages each day, then their vocabulary, on average, is 30 words.

c Yes. When a child reads 0 pages a day, the number of pages is within the data range and the value of the vocabulary makes sense.

d No

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33

a 100

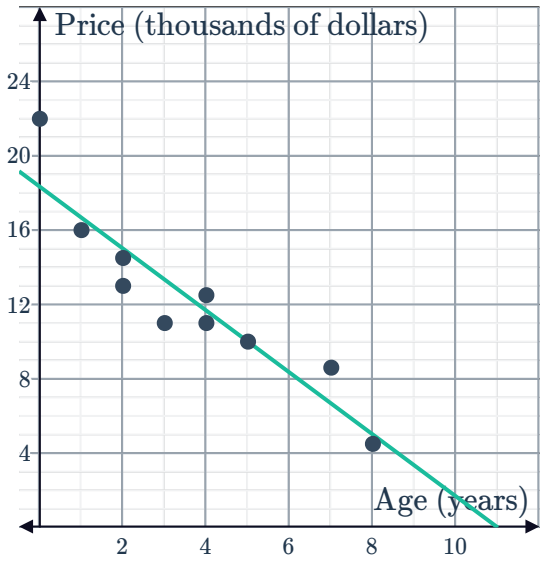
b When a student spends no time using the app, they achieve a math score of 100.

c Yes. When a student spends 0 minutes using the app, their time is within the data range and the value of the score makes sense.

d No

34

a



b $(0, 18\,000)$

c A Toyota Camry that is 0 years old costs $\$18\,000$.

d 11 years

35

a $14\,811$

b The y-intercept indicates that when a house is brand new, its value is, on average, $\$14\,811$.

c No. When a house is zero years old its age is not within the data range.

d Approximately 4.9 years.

e $\$1\,200\,000$

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